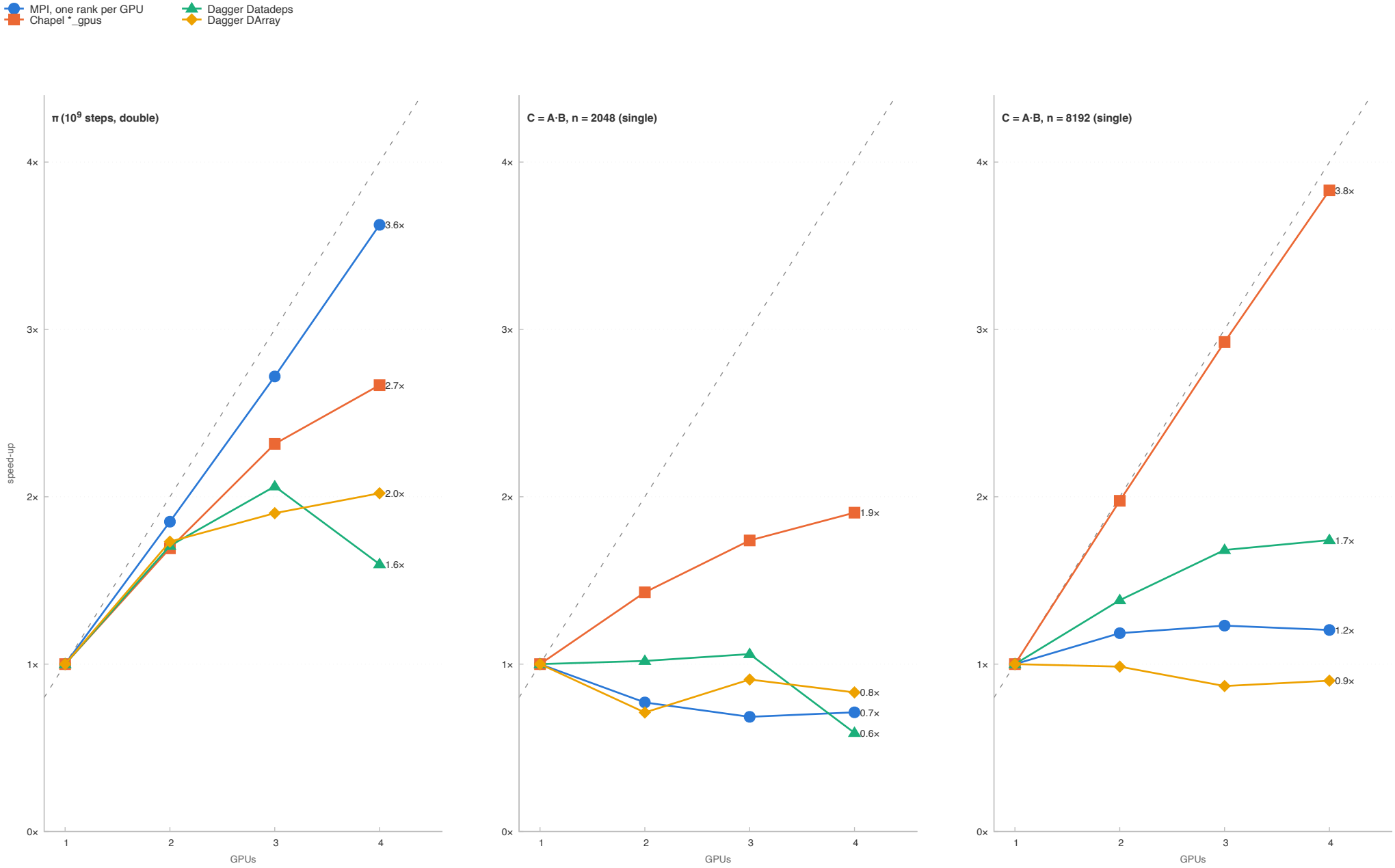
Programs for several GPUs on one to four RTX 3070

Speed-up over the same program on one GPU; the dashed line is the ideal (n GPUs, n times as fast)

4 × NVIDIA GeForce RTX 3070 without peer-to-peer access (g1-4x3070.xlarge, 4 vCPUs), 2026-10-18; median of 3 runs; MPI with one rank per GPU

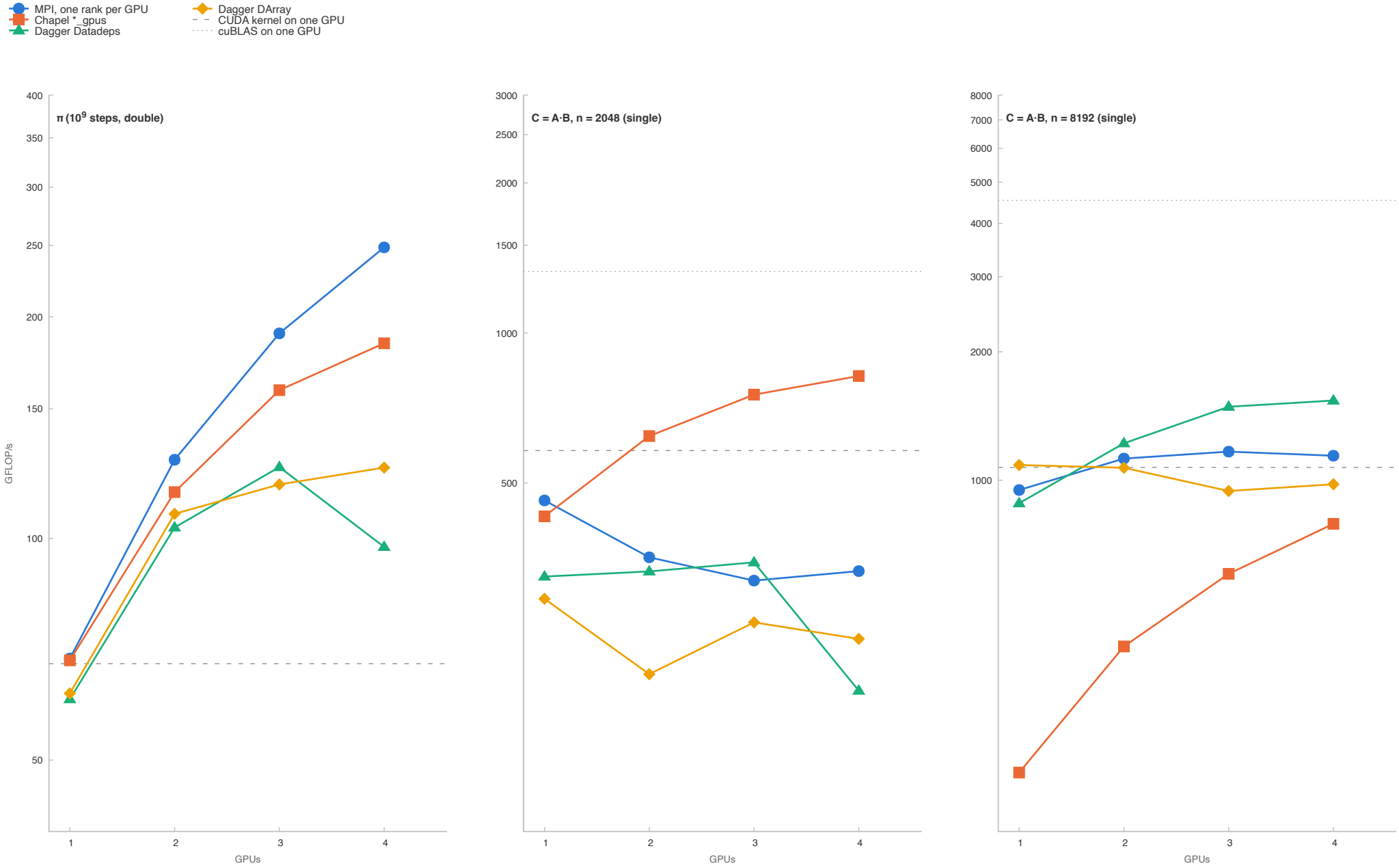


Times include allocating GPU memory and all transfers between host and GPUs; one GPU means CUDA_VISIBLE_DEVICES=0, two GPUs 0,1, and so on.

Programs for several GPUs on one to four RTX 3070

Rate in GFLOP/s on a logarithmic axis; grey lines for comparison: the CUDA kernel (dashed) and cuBLAS (dotted) on one GPU

4 × NVIDIA GeForce RTX 3070 without peer-to-peer access (g1-4x3070.xlarge, 4 vCPUs), 2026-10-18; median of 3 runs; MPI with one rank per GPU



Times include allocating GPU memory and all transfers between host and GPUs; one GPU means CUDA_VISIBLE_DEVICES=0, two GPUs 0,1, and so on.